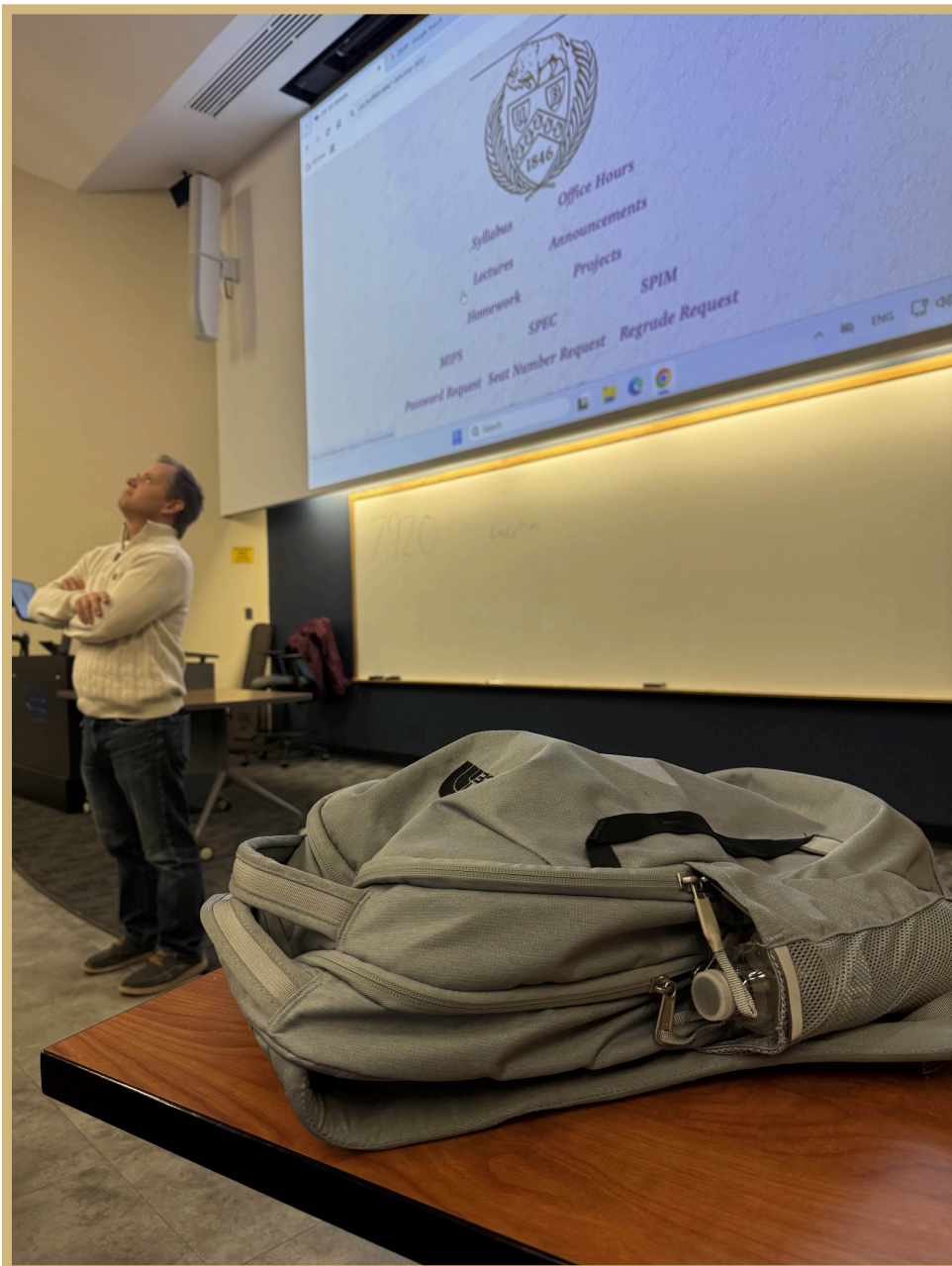


CSE 479: Advanced Embedded Systems – Adaptive Cruise Control

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PROPOSAL



- Our lovely beaver taking a snack break 😊 -

Project Overview

Modern vehicles are increasingly equipped with advanced driver assistance systems that make driving safer and more convenient. One of the most widely adopted technologies is adaptive cruise control, a system that goes beyond traditional cruise control by actively monitoring traffic conditions ahead. Instead of simply maintaining a constant speed, adaptive cruise control uses sensors to detect the vehicle in front and automatically adjusts speed to maintain a safe following distance. When traffic slows down, your car slows down. When traffic speeds up, your car accelerates back to the set cruise speed. This technology has become standard in many new vehicles from manufacturers like Toyota, Honda, Subaru Genesis, Audi and Mazda, and it's basically a stepping stone toward fully autonomous driving systems.

For our CSE 479 final project, we're designing and implementing a scaled down adaptive cruise control system using the TI Tiva C Series EK-TM4C123GXL microcontroller. Our system demonstrates the core principles of adaptive cruise control on a small 4 wheel car platform. The vehicle will autonomously maintain a safe following distance from objects or other vehicles in its path by continuously measuring distance with an ultrasonic sensor and dynamically adjusting motor speed through the microcontroller. The project applies embedded systems concepts like Pulse Width Modulation motor control, Quadrature Encoder Interface for encoder reading, etc.

How It Works

The adaptive cruise control system operates through a continuous sensing, decision making and actuation control loop that runs on the Tiva C microcontroller.

Sensing: The HC-SR04 ultrasonic sensor works like sonar. It shoots out a brief burst of high frequency sound waves and then listens for the echo bouncing back from objects in front of the car. The Tiva C triggers the sensor and measures how long the echo takes to return, then calculates the distance to whatever's ahead.

Decision Making: Once the Tiva C calculates the current distance. For example: it runs through our control algorithm. If the distance is far (greater than 50 cm), the car is in cruise mode and maintains higher speed. If the distance is in the target range (30 to 50 cm), the car enters following mode and adjusts speed to keep a safe gap. If the distance drops below the safe threshold (less than 30 cm), the car activates emergency braking and stops to avoid collision.

Actuation: The Tiva C controls vehicle speed by adjusting the PWM (pulse width modulation) duty cycles sent to the motor driver. PWM is basically a technique where we rapidly switch the motor power on and off. The percentage of time the signal is on versus off determines the average voltage delivered to the motors and therefore their speed. A 100% duty cycle means the motor runs at full speed, 50% means half speed, and 0% means the motors are stopped. Since the Tiva C can't supply enough current directly to the motors, we use TB6612FNG motor driver chips that amplify the control signals and switch the battery power to the motors. The control loop runs continuously at about 10 to 20 times per second so the car responds smoothly to changes in distance.

System Design

Our adaptive cruise control car is built on a Perseids 4 wheel chassis platform that measures 21 cm long by 10 cm wide. The chassis comes with four independent DC gear motors (one for each wheel) which gives us stability, traction, and maneuverability compared to simpler two wheel designs. The motors are rated to operate between 3V and 6V, and we'll power them using four AA batteries providing roughly 6V total.

The Tiva C board sits on the chassis and acts as the main controller. We're using two TB6612FNG dual motor driver modules to control all four motors. Each driver can handle two motors giving us independent control over each wheel. The drivers sit between the Tiva C and the motors, taking low current PWM signals from the Tiva C and switching high current from the battery to power the motors.

The HC-SR04 ultrasonic sensor mounts at the front of the chassis pointing forward to detect obstacles. We are implementing the use of the tachometer encoders that come with each motor. These encoders generate pulses as the motors spin, and we'll use the Tiva C's QEI (Quadrature Encoder Interface) module to read these signals. This is a new concept we'll be learning that lets us measure actual motor speeds and verify our control is working correctly.

We might also add IR infrared obstacle avoidance sensors on the sides or back of the car as extra functionality for detecting obstacles during turns or reversing.

For the control algorithm, we are using a simple distance control gap based strategy. We set target distance thresholds and adjust PWM duty cycle based on which gap the measured distance falls into. The encoders help us make sure all four motors are running at the speeds we command.

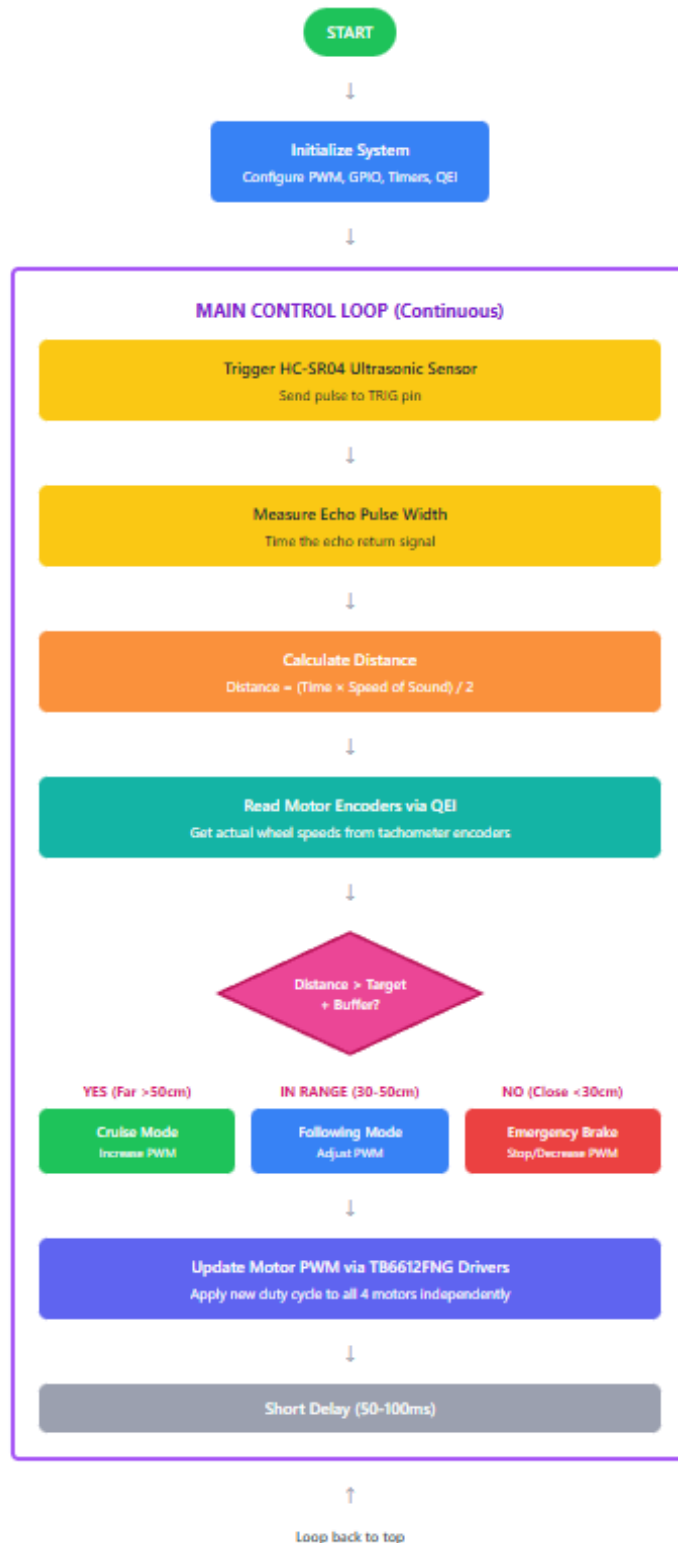
Construction Plan

1. Assemble the chassis kit with motors, wheels, and battery holder
2. Mount the Tiva C on the chassis
3. Connect the two TB6612FNG motor drivers and wire them to all four motors
4. Wire the motor driver control inputs to PWM pins on the Tiva C
5. Mount the HC-SR04 sensor at the front center of the chassis
6. Connect the sensor trigger and echo pins to GPIO pins on the Tiva C
7. Wire the motor encoders to QEI capable pins on the Tiva C
8. Connect the battery pack to power everything
9. Write and upload the control software including sensor reading, distance calculation, PWM motor control, and QEI encoder reading
10. Test and calibrate distance thresholds and motor speeds.

Flow Charts

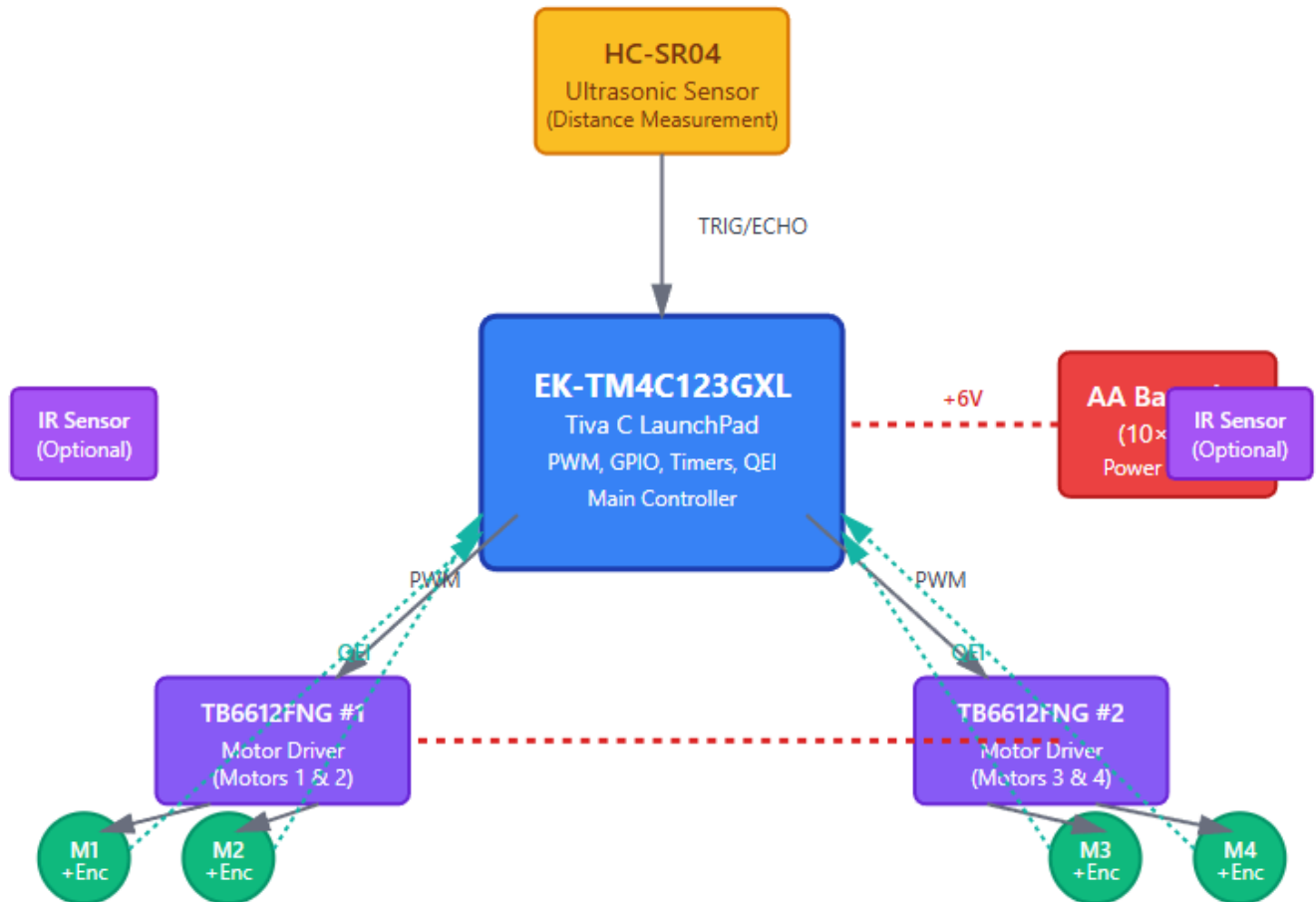
Adaptive Cruise Control System

Control Flow Diagram



System Architecture

System Connection Diagram



Signal Connections:

- HC-SR04 TRIG/ECHO → Tiva C GPIO
- Tiva C PWM → TB6612FNG Motor Drivers (2)
- TB6612FNG → 4× DC Gear Motors
- Motor Encoders → Tiva C QEI pins (teal dashed)
- Optional IR Sensors → Tiva C GPIO

Power Connections:

- AA Batteries → Motor Drivers (Motor Power)
- AA Batteries → Tiva C (Logic Power)
- Common Ground between all components

Bill of Materials – CSE 479 Advanced

Embedded Systems

Item	Purpose	Qty.	Purchase Link	Datasheet Link	Status
Ultrasonic Sensor HC-SR04	Read distance from the forward object	1	Amazon HC-SR04	SparkFun HC-SR04 Data Sheet	Owned & CSE450 Lab
DIY Robot Smart Car Chassis Kit	Provides the Chassis, motors, tachometer encoders, wheels, and battery cover	1	Amazon All-in-One RC Car Frame Kit	No documentation was found :(	Not Received
Amazon Basic AA Alkaline Batteries	Provide Voltage and Current to our components	10	Amazon Basic Batteries	manuals.plus Amazon Basic AA Batteries Manual	Not Received
Dual Motor Driver TB6612FNG	Provides control from the Tiva to the motors without high current damaging the Tiva and other components	2	WWZMDiB TB6612FNG Motor Driver	SparkFun Motor Driver Data Sheet	Not Received
Tiva TM4C123GHPM Microcontroller	The brains to our project and should handle Timers, PWM, SPI, QEI, and all GPIO operations	1	Mouser Link to Tiva Microcontroller	CSE410 Tiva Documentation	WE HAVE HUNDREDS!!!
IR Infrared Obstacle Avoidance Sensor Module	We might use this as an extra functionality to check distance on the sides of our car or maybe even reversing	3	WWZMDiB Infrared Proximity Sensor Module	Handson Technology IR Obstacle Detector Datasheet	In CSE450 Lab